

异质叠层结构孪生体偏差仿真

数学模型数据结构说明书

V2.5 状态传递、局部模型与 N-2-1 过约束配套版

面向 SMS 输入、实测关键特性、测量状态补偿、接触面计算域、结构静力凝聚、界面互补求解、阶段状态/参数传递、界面等效参数、KCP 快速预测、Monte Carlo 与 N-2-1 / 多点过约束装配。

文档类型：硕士论文配套数据结构说明书 / 程序接口依据

版本：V2.5 | 基准格式：V2.2 状态传递与防重复计算配套版 | 日期：2026 年 7 月

重构说明

本版以 V2.2 为底稿样式和章节顺序，保留其“数学形式 - 参数含义 - 数学形式与参数说明”写法；V2.3 的局部模型、样件响应、柔度分离和接触追溯对象，以及 V2.5 的 N-2-1 / 多点过约束对象被嵌入对应正文位置，不再以前置“第一部分/补充块”方式堆叠。

修订记录

版本	定位	主要变化	使用建议
V1.0	接触面计算域数据结构	建立 Part、Interface、ContactDomain、Stage、ContactState、InterfaceParameter 与 KCP 预测对象。	支撑早期程序设计。
V2.0	配套全流程版	补齐八类输入包、实测记录状态字段、参考状态/增量定义、过程实测更新、质量门、缓存和最小运行包。	作为全流程接口依据。
V2.1	公式增强版	在关键数据结构上补充数学形式与参数说明，强化 SMS、过程实测、W_struct/C_n 和 D_validate 边界。	作为公式配套。
V2.2	状态传递与防重复计算配套版	新增 MeasurementStateSnapshot、SMSUpdateTransferPackage、StageState/ParameterSnapshot、StageTransitionRecord、StageIncrement、ContributionLedger 和 DoubleCountCheckResult。	作为主体结构与写法基准。
V2.3	局部模型与离散接触实现配套版	新增局部模型提取、边界继承、样件试验、柔度分离、SMS 离散点映射和 ContactComputationTrace。	嵌入第 2、5、7、9、13、14 章。
V2.5	N-2-1 / 多点过约束配套版	新增扩展候选域、有限刚度约束元素、扩展 LCP/NCP、主动集稳定、接触力非唯一、连接锁定和释放历史对象。	作为技术说明书 V2.4 的现行数据接口依据。

制订说明

本说明书规定数学模型需要哪些数据对象、字段、矩阵、量纲、版本、传递规则和质量标志。技术说明书说明“为什么这样建模、每一步如何执行、假设何时成立”；本说明书强调“程序中必须保存什么、对象之间如何引用、缓存何时失效、质量门如何落地”。

数据结构主线

MeasurementState -> Compensation -> KCM_ref -> SMSPosterior/TransferPackage -> g^0 -> StagePosteriorPackage^(s) -> StageIncrement^(s) -> ContributionLedger -> Δy_{KCP} -> ValidationResult

目录（章节结构与 V2.2 保持一致）

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14	端到端对象流、伪代码与实施清单
附录 A	枚举值、字段命名和最小可运行数据包

层面	技术说明书关注	本数据结构说明书关注	交付物
物理/理论层	原理、假设、简化、适用边界、验证方式。	记录支撑这些原理的数据对象、字段和矩阵维度。	对象字典、字段定义、维度表。
算法/执行层	离线-在线、主动集、参数识别、Monte Carlo、验证闭环。	把算法步骤变成输入、输出、状态、缓存、日志和质量门。	流程接口和伪代码。
实测/校准层	零件级 KCM、过程实测和 KCP 验证的职责边界。	用 support_state、load_state、update_target、data_role 和 DataSplit 防止混用。	实测数据 schema。
工程治理层	适用域、升级触发和模型发布。	记录模型版本、依赖哈希、缓存失效条件和验证结果。	ModelVersionRelease、RunLog。
N-2-1 约束层	说明多点约束接触如何建模、求解和验证。	保存扩展候选域、有限刚度元素、扩展互补解、稳定性和反力非唯一报告。	ExtendedContactDomain、ExtendedLCPsolution 等对象。

- 沿用 V2.2 的 ReferenceState、IncrementDefinition、MeasurementStateSnapshot、SMSUpdateTransferPackage、StagePosteriorPackage、ContributionLedger 和 DoubleCountCheckResult，继续以“参考态 + 阶段增量 + 变形源台账”为主线。
- 纳入 V2.3 的局部模型提取、边界继承、局部 FE/样件试验、位移分解、柔度分离、SMSPointCloudMapping 和 ContactComputationTrace。
- 新增 N-2-1 / 多点过约束对象：OverConstraintLocatorGroup、LocatorElement、LocatorCompliance、ExtendedContactDomain、ExtendedGapAndCompliance、OverConstraintContactModel 和 ExtendedLCPSolution。
- 新增数值治理对象：ActiveSetStabilityTrace、LCPRegularizationConfig、ForceNonuniquenessReport、ConnectionLockHistory 和 ReleaseHistoryRecord。
- 更新最小可运行包和质量门，要求每个过约束求解结果不仅保存 KCP，还保存 q_{ext} 、 W_{ext} 、 λ_{ext} 、 gap_{ext} 、主动集稳定性、反力非唯一和防重复检查。



2 对象分层、全局数据链与八类输入包

2.1 数据对象分层

对象层	典型对象	作用	主要流向
物理实体层	Part、Interface、StageDefinition、KCMDefinition、KCPDefinition	定义研究对象、结合面、阶段和关键特性。	进入 I0、Ikey、Ipred。
几何与形貌层	NominalModel、SMSBasis、SMSPrior、SMSField、SMSUpdateResult	描述名义几何、先验 SMS 和实测更新后的后验 SMS。	进入 I_Gamma 和 KCP。
测量与试验层	MeasurementRecord、MeasurementSet、DataSplit、RegistrationResult	保存实测值、坐标、不确定度和用途划分。	用于 SMS 更新、过程校准和独立验证。
局部模型与样件层	LocalRegionDefinition、LocalSubmodelExtraction、BoundaryInheritanceRecord、InterfaceTestSpecimen、LoadingPath	记录局部模型从全局模型提取、边界继承、样件设计和加载路径。	进入参数识别、柔度分离和适用域。
局部响应与柔度分离层	LocalResponseData、DeformationDecompositionResult、ComplianceSeparationReport	保存总响应、夹具/传感器/基体扣除和 W_struct/C_n 防重复检查。	进入 ParameterLibrary 和质量门。
接触计算层	ContactDomain、ContactPoint、GapField、AreaWeightMatrix、InterfaceLCPModel、ContactComputationTrace	把结合面转化为可求解接触域，并追溯 $g_0 \rightarrow q \rightarrow W \rightarrow \lambda \rightarrow p/w$ 。	进入阶段互补求解和 RunLog。
过约束计算层	ExtendedContactDomain、LocatorElement、LocatorCompliance、OverConstraintContactModel、ExtendedLCPSolution	保存 N-2-1 或多点过约束的扩展候选域、有限刚度元素、扩展互补问题和求解结果。	进入阶段状态、KCP 和验证。
结构凝聚层	DOFPartition、CondensedOperator、ReducedBasis、FactorizationCache、ReducedForceVector	记录 Schur 凝聚、阶段载荷右端项和矩阵缓存。	进入 Ired 和在线求解器。
状态与模式层	LCPSolution、ContactState、ContactMode、StageStateSnapshot、StageTransitionRecord、StageIncrement	保存阶段接触解、模式、阶段状态和阶段增量。	进入界面凝聚和快速预测。
参数与凝聚层	ParameterEstimate、ComplianceDecomposition、InterfaceParameter、ParameterLibrary	保存界面参数、柔度分解、等效误差项和适用域。	进入 KCP 预测和 Monte Carlo。
预测与验证层	KCPPredictionResult、StatisticalPredictionResult、ValidationResult、ModelVersionRelease	保存预测、统计、验证和模型发布结果。	用于论文评价和工程应用。

2.2 八类标准输入包

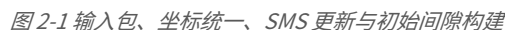
输入包	核心对象	V2.5 增补对象	用途
I0	Part、Interface、NominalModel、Material、AssemblyTopology、SMSPrior	LocalRegionDefinition、N-2-1 约束对象索引	名义模型、局部域定义和过约束对象入口。
I_Gamma	ContactDomain、ContactPoint、G、g0、Bn/Bt、QA、Theta_c0	SMSPointCloudMapping、ExtendedContactDomain、ExtendedGapAndCompliance	接触域、扩展接触域和初始间隙/自由间隙构建。
I_stage	StageInput、BoundaryItem、LoadItem、阶段激活状态、Kr/Fr	OverConstraintLocatorGroup、LocatorElement、ReducedForceVector、ConnectionLockHistory	定位-夹持-连接-释放阶段边界、载荷和过约束历史。
I_key	KCMDefinition、KCPDefinition、FeatureMapping、容差	measurement_type/update_target 的 N-2-1 枚举	定义关键输入/输出和过程观测量。
I_meas	MeasurementPackage、MeasurementRecord、MeasurementSet、DataSplit	InterfaceTestSpecimen、FixtureCalibration、SensorCalibration、LocalResponseData	实测、试验和校准数据治理。

l_red	DOFPartition、CondensedOperator、W_struct、ReducedBasis、FactorizationCache	BoundaryInheritanceRecord、LocalModelValidationResult、material/load dependency	结构凝聚、子模型边界继承和缓存失效。
l_pred	JSSModel、JTTransferModel、ErrorPath、ReferenceState	ContactComputationTrace、ForceNonuniquenessReport、ReleaseHistoryRecord	KCP 传递、反力非唯一和释放回弹贡献追溯。
l_stat	MonteCarloConfig、RandomSample、JointDistribution、ValidationCase	ValidityDomain、局部参数协方差、边界继承不确定度、模式频率	统计预测、敏感性和适用域治理。

参数含义：I_0 为名义模型与先验；I_Gamma 为接触面与扩展接触域；I_stage 为阶段边界、载荷和过约束激活；I_key 为 KCM/KCP 定义；I_meas 为实测与试验包；I_red 为离线结构凝聚包；I_pred 为误差传递与追溯包；I_stat 为统计与验证包。

实测数据更新机制详解：测什么、更新什么、流向哪里

核心原则：实测数据用于约束和校正模型估计，而不是简单替换仿真节点值



3.1 通用元数据 CommonMetadata

参数含义：id/type 标识对象；ver/revision 记录版本；src/source 记录来源；parent 表示依赖对象；Q/quality_flag 表示质量状态；checksum/dependency_hash 用于判断对象是否可追溯、可复用。

3.2 统一编号规则

硕士论文配套数据结构说明书 | 按 V2.2 写法与版式重构

零件	P	P_WALL_L_001	应能识别部位、材料或左右件。
结合面	G	G_RIB03_PANEL_L	建议包含两侧零件或装配区域。
阶段	S	S_CLAMP_02	定位、夹持、连接、释放等阶段。
KCM/KCP	KCM/KCP	KCM_HOLE_A1、KCP_GAP_R03	与特征定义表一一对应。
接触域	CD	CD_G_RIB03_PANEL_L	由 Interface 派生，保存候选点和映射。
扩展过约束域	ECD	ECD_G_RIB03_CLAMP	保存零件接触、定位器、夹持头、孔边连接等扩展候选域。
过约束元素	LOC/CMP/JNT	LOC_PAD_04、CMP_HEAD_02、JNT_HOLE_A3	对应定位、夹持、连接局部对象。
结果	RES	RES_MC_20260704_A	用于预测、统计和验证批次。

3.3 坐标、单位和参考状态

$$\mathbf{x}_{\text{model}} = \mathbf{R} \mathbf{x}_{\text{meas}} + \mathbf{t}; \quad \Delta z = z - z_{\text{ref}}$$

参数含义：R 为旋转矩阵，t 为平移向量；CoordinateSystem 定义坐标基准；ReferenceState 定义某项量相对于哪个状态；IncrementDefinition 定义某一贡献项是绝对量、增量还是残差。

```
CoordinateSystem = {
  coordinate_system_id, name, parent_coordinate_system_id_optional,
  origin[3], rotation_matrix[3,3], handedness,
  definition_source, covariance_optional, metadata
}
ReferenceState = {
  reference_state_id,
  geometry_reference: "nominal | sms_updated | assembled_stage",
  process_reference: "nominal_process | measured_process",
  interface_reference: "nominal_interface | calibrated_interface",
  stage_id_optional, description, locked_fields
}
IncrementDefinition = {
  increment_definition_id, target_quantity, reference_state_id,
  increment_type: "absolute | delta_from_reference | residual | stage_increment",
  included_sources, excluded_sources, unit, sign_convention
}
```

ReferenceState 与 IncrementDefinition 用于说明 SMS 直接贡献、接触压缩贡献、回弹贡献、定位/夹持增量和过程误差贡献相对于哪个基准状态计算，防止同一变形在 KCP 路径中重复计入。

4 实测数据包、用途划分与数据治理

4.1 实测数据包总定义

$$\mathbf{I}_{\text{meas}} = \{\mathbf{I}_{\text{geom}}, \mathbf{I}_{\text{KCM}}, \mathbf{I}_{\text{proc}}, \mathbf{I}_{\text{interface}}, \mathbf{I}_{\text{KCP}}, \mathbf{I}_{\text{split}}\}$$

参数含义：I_geom/I_KCM 用于零件级几何和关键测量；I_proc 用于阶段过程实测；I_interface 用于局部界面试验；I_KCP 用于装配后关键输出验证；I_split 规定识别、校准和验证的数据划分。

```
MeasurementPackage = {
  measurement_package_id,
  geometry_measurement_set_ids, kcm_measurement_set_ids,
  process_measurement_set_ids, interface_test_set_ids, kcp_measurement_set_ids,
  data_split_id, common_coordinate_system_id, unit_system_id,
  preprocessing_pipeline_id, metadata
}
```

4.2 单条实测记录 MeasurementRecord

$$\mathbf{z}_l^{\text{I}}(s) = \mathbf{h}_l^{\text{I}}(s)(\mathbf{x}^{\text{I}}(s), \theta, \mathbf{e}_{\text{proc}}) + \mathbf{v}_l^{\text{I}}(s), \quad \mathbf{v}_l^{\text{I}}(s) \sim \mathbf{N}(\mathbf{0}, \mathbf{R}_l)$$

参数含义： $\mathbf{z}_l^{\text{I}}(s)$ 为第 s 阶段第 l 条实测量； $\mathbf{h}_l^{\text{I}}(s)$ 为观测函数； $\mathbf{x}^{\text{I}}(s)$ 为阶段状态； θ 为界面/工艺参数； $\mathbf{e}_{\text{proc}}$ 为过程误差； $\mathbf{R}_l$ 由 standard_uncertainty/covariance 给出；support_state、load_state、update_target、data_role 决定该数据能更新什么。

```
MeasurementRecord = {
  measurement_id, measurement_set_id, sample_id,
  object_type, object_id, feature_definition_id,
  interface_id_optional, part_id_optional, assembly_id_optional,
```

```

stage_id, support_state, load_state,
timestamp_optional, location, direction, coordinate_system_id,
value, unit, standard_uncertainty, covariance_block_id_optional,
sensor_or_device_id, calibration_certificate_id_optional,
acquisition_method, preprocessing_history,
update_target, data_role, quality_flag,
reference_state_id_optional, increment_definition_id_optional,
missing_reason_optional, metadata
}

```

字段	推荐枚举/形式	作用
support_state	FREE、KNOWN_SUPPORT、FIXTURED、UNKNOWN	说明零件是否处于自由/规定支承状态，是 SMS 更新能否成立的关键。
load_state	FREE、LOCATED、CLAMPED、JOINED、RELEASED、DISASSEMBLED_FREE	区分装配前、过程中和装配后测量。
update_target	SMS、POSE、STAGE_STATE、PROCESS_LOAD、INTERFACE_PARAMETER、REBOUND_PARAMETER、KCP_VALIDATION、OVERCONSTRAINT_STATE、JOINT_LOCK_HISTORY、RELEASE_HISTORY	明确该测量允许更新的模型对象。
data_role	IDENTIFY、CALIBRATE、VALIDATE、MONITOR、EXCLUDE	防止同一数据既调参又验证。
measurement_type	point_cloud、KCM、force、gap、strain、KCP、LOCATOR_REACTION、CLAMP_HEAD_DISPLACEMENT、FASTENER_PRELOAD、RELEASE_REBOUND	描述观测物理量。
quality_flag	PASS、WARN、REJECT、OUTLIER_REVIEWED、MISSING	控制数据是否允许进入计算。

4.3 数据划分 DataSplit

$D_{\text{meas}} = D_{\text{identify}} \cup D_{\text{calibrate}} \cup D_{\text{validate}}, \quad D_{\text{validate}} \cap (D_{\text{identify}} \cup D_{\text{calibrate}}) = \text{empty}$

参数含义：D_identify 用于参数识别；D_calibrate 用于模型校准和超参数选择；D_validate 只用于独立验证；grouping_constraint 防止同一零件/工况泄漏到不同集合。

```

DataSplit = {
    data_split_id,
    identification_sample_ids, calibration_sample_ids, validation_sample_ids,
    split_strategy, grouping_constraint, leakage_check_result,
    random_seed_optional, metadata
}

```



图 4-1 实测数据更新机制：零件级、过程级和装配后数据的职责分离

5 SMS、KCM 更新与初始间隙构建数据结构

5.1 零件、名义模型与 SMS 对象

$$P = \{P_i\}_{i=1}^n, \quad X_i = X_i^0 + \delta_i^{\text{SMS}}, \quad \delta_i^{\text{SMS}} = N_i \alpha_i$$

参数含义：P_i 为零件；X_i^0 为名义几何；delta_i^SMS 为装配前制造形貌偏差；N_i 为形貌基；alpha_i 为低维形貌参数；SMSField 保存形貌场，SMSBasis 保存基矩阵和模态信息。

```
Part = {
  part_id, part_name, material_type, role_in_assembly,
  nominal_model_id, sms_prior_id, material_id,
  thickness_definition_id, layup_id_optional,
  local_coordinate_system_id, mesh_or_point_set_id,
  key_interface_ids, related_kcm_ids, related_kcp_ids, metadata
}
SMSField = {
  sms_field_id, part_id, representation_type,
  reference_nominal_model_id, coordinate_system_id,
  value_location_type, values, direction_type,
  basis_id_optional, alpha_optional, covariance_optional,
  support_state, gravity_compensation_state,
  source_measurement_set_id_optional, reconstruction_error,
  valid_region, metadata
}
SMSBasis = {
  basis_id, part_id, basis_type, N,
  mode_names, mode_units, orthogonality_info,
  training_sample_ids, retained_energy_ratio_optional,
  boundary_constraints, observable_kcm_ids, metadata
}
```

5.2 KCM 定义和 SMS 更新对象

$$m_{\text{KCM}} = H_m N \alpha + b_m + \epsilon_m; \quad \alpha_{\text{hat}} = \operatorname{argmin}_{\alpha} ||m_{\text{KCM}} - H_m N \alpha - b_m||^2_{R_m^{-1}} + \lambda ||L(\alpha - \alpha_0)||^2$$

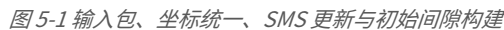
参数含义：m_KCM 为关键测量特性；H_m 为观测矩阵；b_m 为偏置；R_m 为测量协方差；L/lambda 为正则化约束；alpha_0/alpha_hat 分别为先验和后验 SMS 参数；SMSUpdateResult 保存更新结果和质量指标。

```
KCMDefinition = {
  kcm_id, kcm_name, feature_type,
  target_object_type, target_object_id,
  stage_id_optional, nominal_location, evaluation_direction,
  extraction_or_measurement_matrix_id,
  nominal_value, expected_unit, measurement_method,
  required_uncertainty, update_target, metadata
}
SMSUpdateConfig = {
  update_config_id, sms_prior_id, sms_basis_id,
  kcm_measurement_set_ids, H_m_id, R_m_id,
  regularization_type, regularization_matrix_L,
  lambda_value, lambda_selection_method,
  constraints_optional, solver_name, convergence_tolerance,
  data_role_allowed: ["IDENTIFY", "CALIBRATE"], metadata
}
SMSUpdateResult = {
  update_result_id, update_config_id,
  alpha_prior, covariance_prior,
  alpha_posterior, covariance_posterior,
  reconstructed_sms_field_id, predicted_kcm, residual_kcm,
  weighted_residual_norm, regularization_norm,
  condition_number, observability_rank,
  posterior_variance_reduction,
  active_constraints, quality_flag, metadata
}
```

5.3 初始间隙构建对象

$$g^0 = g_{\text{nom}} + G \delta_{\text{hat}}^{\text{SMS}} + G_{\text{pose}} e_{\text{pose}}; \quad g_k^0 = n_k^T (x_{\{j,k\}} - x_{\{i,k\}})$$

GapField 必须记录 nominal_component 与 sms_component, 这样后续 KCP 预测中可以区分“装配前几何直接贡献”和“接触压缩附加贡献”, 避免重复计入。


$$\mathbf{m_raw}^{(m)} = H_{\text{delta}} N \alpha + H_{\text{pose}} \mathbf{e_pose}^{(m)} + H_u \mathbf{u_struct}^{(m)} + H_c \mathbf{w_local}^{(m)} + \mathbf{b_m} + \epsilon_m$$

```
MeasurementStateSnapshot = {
    measurement_state_id, measurement_set_id, sample_id, part_id,
    support_state, load_state, target_reference_state_id,
    boundary_condition_ids, measured_load_ids, fixture_model_id,
    gravity_vector, temperature_optional,
    measurement_coordinate_system_id, model_coordinate_system_id,
    coordinate_transform_id, transform_covariance_optional,
    acquisition_timestamp, state_quality_flag, metadata
}

MeasurementDeformationEstimate = {
    deformation_estimate_id, measurement_state_id,
    pose_estimate, pose_covariance,
    structural_displacement_field_id,
    local_compression_field_id_optional,
    fixture_deflection_field_id_optional,
    compensation_vector_in_kcm_space,
    compensation_covariance,
    included_deformation_source_ids,
    estimation_method, source_model_ids,
}
```

```
residual_and_identifiability, quality_flag, metadata
}
```

字段	规则
support_state / load_state	必须与目标 reference_state 可转换；UNKNOWN 状态不得直接更新 SMS。
boundary/load/fixture IDs	任一依赖变化，补偿缓存和 SMS 后验必须失效。
included_deformation_source_ids	为防重复台账提供唯一变形源标识，如 GRAVITY_M、FIXTURE_M、LOCAL_CONTACT_M。
estimation_method	FE_CORRECTION、MULTI_POSE_SEPARATION、STATE_DIFFERENCE 或 NONE_REJECT。

5.5 等效参考态 KCM、SMS 后验与首阶段传递包

$$m_ref = m_raw - c_pose - c_struct - c_local - b_m; \quad R_ref = R_raw + R_comp + R_transform$$

参数含义：m_ref 为补偿后的参考态 KCM；c_pose、c_struct、c_local 分别为位姿、结构变形和局部接触补偿；R_ref 为合并原始测量、补偿和坐标转换不确定度后的等效协方差。

```
MeasurementStateCompensationResult = {
  compensation_result_id, measurement_state_id,
  raw_kcm_values, raw_covariance_R,
  pose_component, structural_component,
  local_contact_component_optional, bias_component,
  equivalent_reference_kcm, equivalent_covariance_R_ref,
  compensation_method, target_reference_state_id,
  excluded_deformation_source_ids,
  identifiability_metrics, quality_flag, metadata
}
SMSUpdateTransferPackage = {
  sms_transfer_package_id, part_id, sample_id,
  sms_update_result_id, reconstructed_sms_field_id,
  alpha_posterior, covariance_posterior,
  target_reference_state_id,
  compensation_result_id,
  included_source_ids, excluded_source_ids,
  valid_for_stage_ids, gap_mapping_version,
  package_hash, quality_flag, metadata
}
SMSUpdateConfig += {
  measurement_state_snapshot_id, compensation_result_id,
  target_reference_state_id, allowed_compensation_methods,
  forbidden_deformation_source_ids
}
SMSUpdateResult += {
  reference_state_id, compensation_trace_ids,
  included_source_ids, excluded_source_ids,
  transfer_package_id, posterior_cross_covariance_optional
}
GapField += {
  source_sms_transfer_package_ids,
  excluded_measurement_deformation_source_ids,
  double_count_check_id
}
```

5.6 SMS 离散点到接触计算追溯对象

$$g^0 = g_nom + G \delta_SMS + G_pose e_pose$$

参数含义：g^0 为候选接触点上的初始间隙；delta_SMS 可以来自点云、模态系数、FE 节点值或法向残差；G 负责插值/投影；G_pose e_pose 表示装配初始位姿偏置对间隙的影响。新增对象用于追溯 SMS 点或低维模态如何被投影到接触候选点，并与后续 q、W、lambda、p、w 的计算链关联。

```
SMSPointCloudMapping = {
  mapping_id, part_id, sms_field_id, contact_domain_id,
  sms_representation_type: "point_cloud | modal | FE_node | surface_normal",
  source_point_ids_or_mode_ids,
  target_contact_point_ids,
  interpolation_method,
  projection_direction_rule,
  G_gap_mapping_id,
  mapping_residual_statistics,
}
```

```

    validity_region, metadata
}
ContactComputationTrace = {
    trace_id, sample_id, stage_id, contact_domain_id,
    sms_transfer_package_ids,
    gap_field_id, stage_input_id, condensed_operator_id,
    q_vector_id, W_struct_id, Cn_local_id, W_total_id,
    lcp_solution_id,
    pressure_field_id, local_compression_field_id,
    cache_ids_used,
    double_count_check_id,
    quality_flag, metadata
}

```

字段	规则
sms_representation_type	必须说明 SMS 是点云、模态系数、FE 节点值还是法向残差。
projection_direction_rule	必须与 ContactPoint.normal_n 一致；法向改变时 G 和 GapField 失效。
mapping_residual_statistics	用于判断 SMS 插值/投影是否会影响接触集或 KCP。
ContactComputationTrace	每个阶段求解至少保存一次，用于回放 $g_0 \rightarrow q \rightarrow W \rightarrow \lambda \rightarrow p/w$ 的计算链。

6 接触面计算域与阶段输入数据结构

6.1 接触面计算域 ContactDomain

$$I_{\{\Gamma a\}} = \{\Omega_a, x_i^0, x_j^0, n_a, T_a, Q_A, G_a, B_{\{n,a\}}, B_{\{t,a\}}, \Theta_{\{c,a\}}^0\};$$

$$\lambda_{n} = Q_A p_n$$

参数含义： Ω_a 为候选接触点集合； n/T 为法向和切向基； Q_A 为面积权重； G 映射 SMS 到初始间隙； B_n/B_t 映射结构位移到法向/切向相对运动； Θ_c^0 为初始接触参数。

```

ContactDomain = {
    contact_domain_id, interface_id,
    side_i_region_id, side_j_region_id,
    contact_point_ids,
    correspondence_method, normal_definition_method,
    tangent_definition_method,
    area_weight_matrix_id,
    G_gap_mapping_id, Bn_mapping_id, Bt_mapping_id_optional,
    nominal_gap_field_id, initial_contact_parameter_id,
    candidate_region_margin, density_level,
    validity_region, metadata
}
ContactPoint = {
    point_id, contact_domain_id, local_index,
    x_i0, x_j0, normal_n, tangent_basis_T,
    area_weight, local_coordinate_system_id,
    candidate_flag, edge_or_interior_flag,
    correspondence_quality, metadata
}

```

对象	推荐维度	说明
G_gap_mapping	$m \times n_{\text{geo}}$ 或 $m \times r_{\text{SMS}}$	SMS 到法向初始间隙的几何映射。
Bn_mapping	$m \times n_{\Gamma}$	界面自由度到法向闭合量的映射。
Bt_mapping	$2m \times n_{\Gamma}$	界面自由度到切向相对位移的映射，可选。
AreaWeightMatrix Q_A	$m \times m_a$	候选点面积权重，用于力/压力转换和面积积分。
InitialContactParameter Θ_{c0}	按界面/阶段索引	初始或预识别 C_n 、 C_t 、 μ 、阈值等。

6.2 阶段输入 StageInput、BoundaryItem 与 LoadItem

$$I_{\text{stage}}(s) = \{B(s), L(s), K_r(s), F_r(s), A_{\text{act}}(s), S_0(s)\}$$

参数含义： $B(s)$ 为阶段边界约束； $L(s)$ 为夹持/预紧/连接载荷； $K_r(s)$ 、 $F_r(s)$ 为消除规定边界后的阶段刚度与载荷； $A_{\text{act}}(s)$ 为约束/载荷/连接激活状态； $S_0(s)$ 为上一阶段继承状态。

```

StageInput = {
    stage_input_id, stage_id,

```

```

boundary_item_ids, load_item_ids, active_joint_ids,
constraint_activation_vector, load_activation_vector, joint_activation_vector,
reduced_stiffness_id, reduced_force_vector_id,
inherited_state_id, expected_measurement_set_ids,
reference_state_id, metadata
}
BoundaryItem = {
    boundary_id, boundary_type, target_object_id,
    target_dofs, direction, prescribed_value,
    stiffness_optional, stage_activation, uncertainty_optional,
    measured_value_id_optional, metadata
}
LoadItem = {
    load_id, load_type, target_object_id,
    direction, magnitude, load_curve_optional,
    distributed_area_optional, stage_activation,
    uncertainty_model_id_optional, measured_value_id_optional,
    metadata
}

```

阶段	主要边界/载荷	推荐实测	优先更新对象
定位	定位基准、销/支承约束、自重	定位点位移、初始局部间隙、相对位姿	e_pose、夹具零偏、初始接触集
夹持	定位 + 夹持力/位移	夹持力、夹持头位移、局部间隙、应变	F_clamp、S^(2)、必要时 Cn
连接	夹持 + 预紧/连接锁定	预紧力、孔位、连接区应变、切向滑移	F_preload、连接状态、Ct/mu
释放	保留连接，撤除部分夹持/定位	前后位移、最终间隙/阶差/回弹	S^(4)、beta_r；验证工况只评价

7 离线结构凝聚 I_red 与缓存数据结构

7.1 自由度分区与凝聚算子

$$[K_{II} \ K_{I\Gamma}; K_{\Gamma I} \ K_{\Gamma\Gamma}][u_I; u_\Gamma] = [F_I; F_\Gamma], \quad K_\Gamma = K_{\Gamma\Gamma} - K_{\Gamma I} K_{II}^{-1} K_{I\Gamma}, \quad W_{struct} = B_\Gamma K_\Gamma^{-1} B_\Gamma^T$$

参数含义：u_I 为内部自由度，u_Gamma 为界面/输出相关自由度；K_Gamma 为 Schur 凝聚刚度；W_struct 为结构贡献的界面柔度；DOFPartition 保存自由度分区，CondensedOperator 保存凝聚结果。

```

DOFPartition = {
    dof_partition_id, stage_id,
    full_free_dof_ids, internal_dof_ids, interface_dof_ids,
    kcp_related_dof_ids, prescribed_dof_ids,
    permutation_vector, rigid_body_mode_check, metadata
}
CondensedOperator = {
    condensed_operator_id, stage_id, dof_partition_id,
    K_II_id, K_I_Gamma_id, K_Gamma_I_id, K_Gamma_Gamma_id,
    Kbar_Gamma, Fbar_Gamma,
    W_struct_id,
    factorization_id,
    symmetry_error, positive_definiteness_check,
    condition_estimate, derivation_source_model_id,
    included_physics_struct, metadata
}

```

included_physics_struct 应说明 W_struct 已经包含哪些结构响应，例如零件基体弯曲、整体柔性、连接锁定或边界弹性；这将作为防重复柔度判断的依据。

7.2 缓存与失效规则

$$F_{cache} = \text{factor}(K_{\Gamma\Gamma}) \text{ or } \text{factor}(K_{II}), \quad \text{reuse} \Leftrightarrow \text{hash}(K, B, \text{mesh}, \text{material}, \text{boundary}) \text{ unchanged}$$

参数含义：factorization_type 为分解类型；matrix_checksum 和 reusable_conditions 判断缓存可复用性；CacheInvalidationRule 定义当几何、网格、材料、边界或映射变化时的重建/回算动作。

```

FactorizationCache = {
    factorization_id, matrix_id, factorization_type,
    permutation, factor_data_reference,

```



```
matrix_checksum, software_version,
reusable_conditions, invalidation_rules, metadata
}
CacheInvalidationRule = {
    rule_id, target_cache_type,
    dependency_fields,
    invalidation_condition,
    required_action: "reuse | rebuild_operator | high_fidelity_recompute | reject_prediction",
    severity: "INFO | WARN | FAIL"
}
```

7.3 材料、边界与载荷缓存依赖

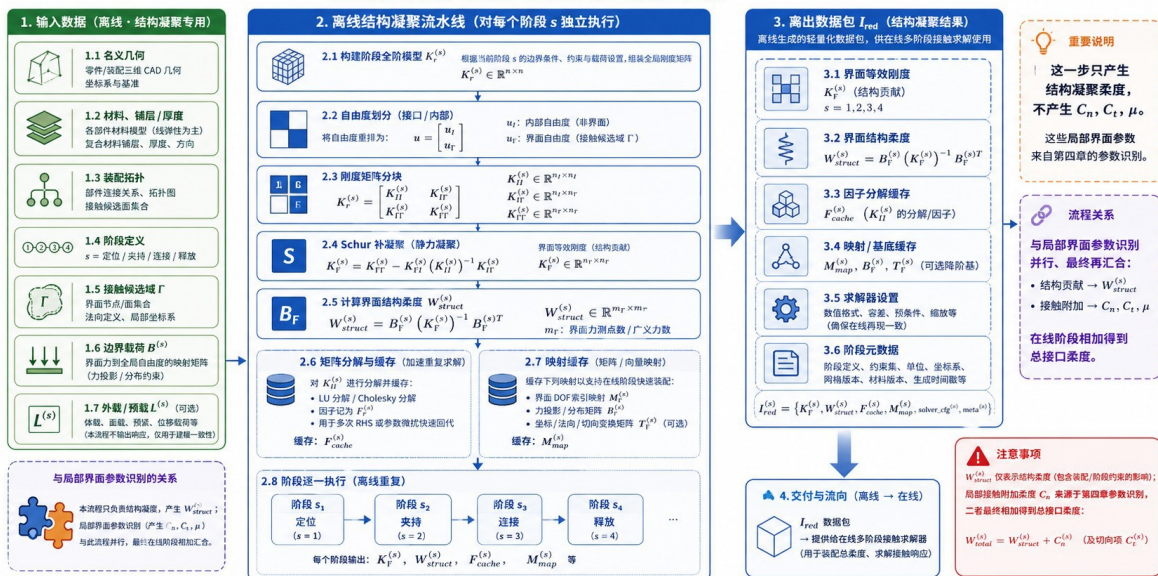
CondensedOperator 进一步记录材料、铺层、厚度、连接和阶段边界依赖，保证材料/外载改变时缓存失效规则明确。

```
CondensedOperator += {
    material_dependency_ids,
    layup_dependency_ids_optional,
    thickness_dependency_ids,
    joint_state_dependency_ids,
    boundary_condition_dependency_ids,
    load_reference_ids,
    valid_stage_load_range,
    valid_geometry_deviation_range,
    operator_rebuild_rule,
    metadata
}
ReducedForceVector = {
    reduced_force_vector_id, stage_id,
    source_load_item_ids,
    gravity_component,
    clamp_component,
    preload_component,
    prescribed_displacement_equivalent_component_optional,
    force_covariance_optional,
    unit, metadata
}
```

变化对象	失效对象	required_action
material_dependency_ids / layup_dependency_ids	K_Gamma、W_struct、FactorizationCache	REBUILD_IRED
load_reference_ids	F_Gamma、u_Gamma、F、q	RESOLVE_LCP
boundary_condition_dependency_ids	K_Gamma、F_Gamma、q、W_struct	REBUILD_IRED 或 RESOLVE_LCP
Cn_local_id	W_total、W_CC 分解、LCP 模式	REBUILD_W
G_gap_mapping_id / normal_definition_method	GapField、q、ContactMode	REBUILD_G0 + RESOLVE_LCP

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离线结构静力凝聚细化图

从阶段全阶段模型到界面结构柔度 W_{struct} 与 I_{red} 

8 互补求解、接触模式与阶段状态数据结构

8.1 InterfaceLCPModel 与求解配置

$$\begin{aligned} q^\wedge(s) &= g^\wedge\theta - B_Gamma \ u_{{Gamma},F}^\wedge(s), & W^\wedge(s) &= W_struct^\wedge(s) + C_n^\wedge(s), & 0 &\leq \lambda_n^\wedge(s) \perp \\ g^\wedge(s) &= q^\wedge(s) + W^\wedge(s) \lambda_n^\wedge(s) \geq 0 \end{aligned}$$

参数含义：q 为自由间隙；W 为总界面柔度； λ_n 为法向接触广义力；g 为平衡后间隙；InterfaceLCPModel 保存 LCP 矩阵和未知量定义，SolverConfig 保存主动集/互补求解参数。

```
InterfaceLCPModel = {
    lcp_model_id, sample_id, stage_id,
    contact_domain_ids,
    q_mapping_components,
    W_struct, Cn_local, W_total,
    area_weight_matrix_id,
    unknown_type: "generalized_contact_force",
    gap_sign_convention,
    complementarity_tolerances,
    solver_config_id,
    validity_region, metadata
}
ComplementaritySolverConfig = {
    solver_config_id, solver_type,
    active_set_initialization,
    gap_tolerance, force_tolerance, complementarity_tolerance,
    maximum_iterations, scaling_rule,
    line_search_or_regularization_optional,
    warm_start_enabled, fallback_solver_optional,
    metadata
}
```

8.2 LCPSolution、ContactMode 与 ContactState

$$S^\wedge(s) = \{C, D, g, \lambda_n, p_n, u_Gamma, w_n, d_t, \Delta r, \chi\}, \quad p_n = Q_A^{-1} \lambda_n, \quad w_n = C_n \lambda_n$$

参数含义：C/D 为接触/分离集合； χ 为接触模式； p_n 为压力； w_n 为局部压缩； d_t 为切向位移； Δr 为回弹；LCPSolution 保存单次求解，ContactState 保存阶段状态及继承关系。

```
LCPSolution = {
    solution_id, lcp_model_id, sample_id, stage_id,
    lambda_n, pressure_p_n, gap_g,
    active_contact_indices, inactive_indices,
    equilibrium_residual, gap_violation, force_violation,
    complementarity_residual, iteration_count,
    convergence_status, warm_start_source_id_optional,
    metadata
}
ContactMode = {
    contact_mode_id, stage_id,
    active_contact_indices, stick_indices, slip_indices,
    mode_hash, feasible_region_description,
    representative_sample_ids, occurrence_count, metadata
}
ContactState = {
    contact_state_id, sample_id, stage_id,
    interface_state_ids, global_or_interface_displacement,
    equilibrium_residual,
    inherited_from_state_id_optional,
    state_update_source_measurement_ids,
    convergence_summary, quality_flag, metadata
}
```

多阶段主动集与互补求解细化图

定位—夹持—连接—释放的阶段循环与状态继承

图例： □ 输入/输出/状态 □ 处理/计算步骤
✖ 迭代求解相关 ✖ 状态继承/传递
→ 不可行/反馈迭代



图 8-1 多阶段主动集与互补求解流程

8.3 阶段状态、参数快照与阶段传递对象

$$x_s^- = \Phi_s(x_{s-1}^+, \Delta \hat{x}_{SMS}, c_s, \theta_s^-); \quad (x_s^+, \theta_s^+) = U_s(x_s^-, \theta_s^-, z_s)$$

参数含义: x_s^- 为第 s 阶段预测状态; x_s^+ 为融合实测后的后验状态; θ_s 为阶段或界面参数; c_s 为阶段控制输入; z_s 为阶段过程观测。

```
StageStateSnapshot = {
  stage_state_snapshot_id, sample_id, stage_id,
  state_role: "PREDICTED | POSTERIOR",
  reference_state_id, parent_stage_state_id_optional,
  pose_state, interface_displacement,
  gap_g, lambda_n, lambda_t_optional,
  pressure_p_n, local_compression_w_n,
  tangential_slip_d_t_optional, contact_mode_id,
  joint_lock_state_optional, state_covariance_id_optional,
  absolute_or_increment, source_ledger_id,
  quality_flag, metadata
}

StageParameterSnapshot = {
  stage_parameter_snapshot_id, stage_id, interface_id_optional,
  parameter_names, parameter_values,
  covariance, parameter_scope,
  persistence_policy, validity_domain_id,
  source_prior_or_update_ids,
  update_permission, active_bounds,
  quality_flag, metadata
}

StageTransitionRecord = {
  transition_id, from_stage_id, to_stage_id,
  from_state_id, to_predicted_state_id,
  state_transfer_matrix_or_function,
  parameter_transfer_rules,
  activated_boundary_ids, deactivated_boundary_ids,
  activated_load_ids, removed_load_ids,
  actual_load_increment, joint_activation_change,
  locked_history_ids, covariance_transfer_id,
  dependency_hash, quality_flag, metadata
}
```

8.4 阶段增量、协方差传递与后验包

$$\Delta x^+(s) = x_s^+ - T_{\{s-1 \rightarrow s\}} x_{s-1}^+; \quad P_{\{s+1\}}^- = F_s P_s^+ F_s^T + Q_s$$

参数含义： $\Delta x^{(s)}$ 为阶段新增状态增量； $T_{\{s-1 \rightarrow s\}}$ 为阶段间状态传递； P 为状态协方差； F_s 为阶段状态雅可比； Q_s 为过程噪声。

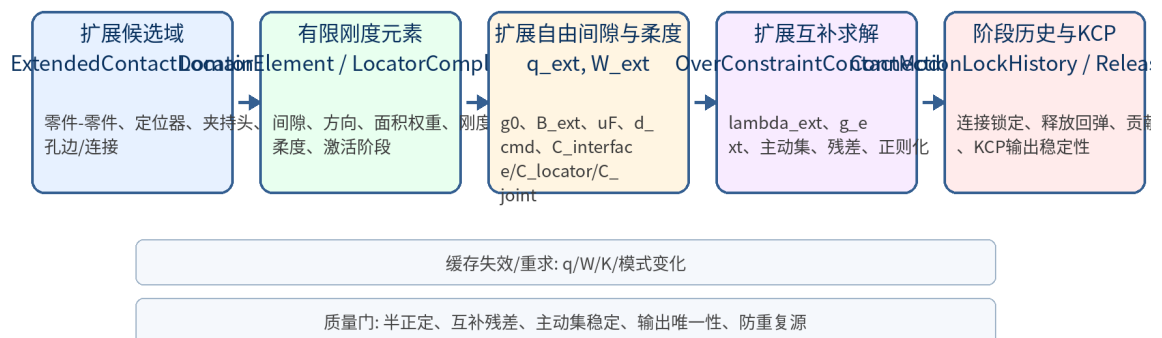
```
StageIncrement = {
  stage_increment_id, sample_id, stage_id,
  from_reference_state_id, to_reference_state_id,
  source_state_id, target_state_id,
  increment_vector, increment_components,
  increment_definition_id,
  deformation_source_ids,
  covariance_optional,
  consumed_by_ids, quality_flag, metadata
}
StageCovarianceTransfer = {
  covariance_transfer_id, transition_id,
  state_jacobian_F, parameter_jacobian_optional,
  process_noise_Q, cross_covariance_rule,
  prior_covariance_id, posterior_covariance_id,
  approximation_validity, metadata
}
StagePosteriorPackage = {
  stage_posterior_package_id, sample_id, stage_id,
  posterior_state_snapshot_id,
  posterior_parameter_snapshot_ids,
  stage_increment_id, transition_record_id,
  locked_history_ids, source_ledger_id,
  parent_package_id_optional,
  package_hash, quality_flag, metadata
}
```

阶段	主要更新	持续传递	不得传递/重复
LOCATE	pose、fixture bias、locating reaction、initial contact	定位后几何、活动约束、模式暖启动、 P	不写回 SMS；不凭单点更新 C_n
CLAMP	actual clamp load、state、必要时 C_n	夹持后状态、实际载荷、接触历史、参数后验	不再次加入测量夹持补偿
JOIN	preload、 k_{joint} 、lock、slip、必要时 C_t/μ	locked_reference、joint state、slip history	不把几何重置为 SMS
RELEASE	final state、rebound；校准时 β_r/R_r	最终状态、KCP 增量和质量标志	不同时累加 JOIN 与 RELEASE 绝对状态

8.5 N-2-1 / 多点过约束对象与扩展互补数据结构

本节与技术说明书中的 N-2-1 / 多点过约束接触状态计算统一模型——对应，集中定义 N-2-1 / 多点过约束计算所需对象。写法沿用 V2.2 的“数学形式、参数含义、数学形式与参数说明”。

N-2-1 / 多点过约束数据对象闭环



与 V2.2 的 StagePosteriorPackage、ContributionLedger、QualityGate 保持同一对象流

图 8-2 N-2-1 / 多点过约束数据对象闭环

8.5.1 扩展候选域对象

$$I_ext^s = \{I_Gamma^s, I_locator^s, I_clamp^s, I_joint^s, M_ext^s\}$$

参数含义： I_ext^s 表示第 s 阶段扩展候选接触域； I_Gamma^s 为零件-零件结合面域； $I_locator^s$ 为定位器、支承点、卡板线/面等工装约束域； I_clamp^s 为夹持头、压头或强迫位移接触域； I_joint^s 为孔边、紧固件承压、预紧和连接锁定域； M_ext^s 为扩展域内点、线、面、孔边和连接自由度之间的映射关系。

```
ExtendedContactDomain = {
    extended_domain_id, stage_id,
    part_contact_domain_ids, locator_domain_ids, clamp_domain_ids, joint_domain_ids,
    extended_contact_point_ids,
    extended_gap_mapping_id, extended_B_mapping_id,
    locator_group_ids, validity_region, quality_flag, metadata
}
```

8.5.2 有限刚度定位/夹持/连接元素

$$e_j = (type_j, x_j, n_j, T_j, A_j, C_j, g_j^0, d_cmd_j, b_j, s_active_j)$$

参数含义： e_j 为第 j 个过约束元素； $type_j$ 表示 LOCATOR_POINT、SUPPORT_PAD、CARD_LINE、CLAMP_HEAD、FASTENER_BEARING 或 JOINT_LOCK； x_j 为作用位置； n_j/T_j 为法向与切向基； A_j 为面积权重； C_j 为该元素局部柔度或等效柔度矩阵； g_j^0 为初始间隙； d_cmd_j 为命令位移、预紧沉降或强迫闭合量； b_j 为单边/双边属性； s_active_j 为激活阶段。

```
OverConstraintLocatorGroup = {
    locator_group_id, stage_ids, locator_element_ids,
    group_type: "N21 | MULTI_POINT | CARD | CLAMP | JOINT",
    activation_rule, reference_state_id, uncertainty_model_id,
    quality_flag, metadata
}
LocatorElement = {
    locator_element_id, locator_group_id, element_type, target_object_id,
    location, normal_n, tangent_basis_T, area_weight,
    unilateral_or_bilateral, initial_clearance_g0, command_displacement,
    preload_optional, active_stage_ids, source_ids, metadata
}
LocatorCompliance = {
    compliance_id, locator_element_id, stiffness_or_compliance_type,
    normal_compliance_Cn, tangential_compliance_Ct_optional,
    covariance_optional, validity_domain_id,
    source_calibration_ids, quality_flag, metadata
}
```

字段说明：OverConstraintLocatorGroup 管理一组定位、卡板、夹持或连接元素；LocatorElement 描述几何、方向、面积权重、单边/双边属性、初始间隙和命令位移；LocatorCompliance 保存有限刚度/柔度及其不确定度。 C_j 不应与 W_struct 中已包含的结构柔度重复。

8.5.3 扩展自由间隙与扩展柔度

$$q_ext^s = g_ext^0 - B_ext^s u_F^s - d_cmd^s$$

参数含义： q_ext^s 为第 s 阶段无接触反力时的扩展自由间隙； g_ext^0 为扩展候选域初始间隙； B_ext^s 为结构位移到扩展闭合量的映射； u_F^s 为自由外载、重力、夹持力和预紧力产生的自由结构位移； d_cmd^s 为夹持头命令位移、卡板压紧量或连接预沉降向量。

$$W_ext^s = W_struct_ext^s + C_interface^s + C_locator^s + C_joint^s$$

参数含义： W_ext^s 为扩展总柔度； $W_struct_ext^s = B_ext(K_Gamma)^{-1} B_ext^T$ 为凝聚结构柔度在扩展域上的投影； $C_interface^s$ 为零件结合面局部附加柔度； $C_locator^s$ 为定位器、卡板、夹持头等局部柔度； C_joint^s 为紧固件、孔边承压和连接锁定局部柔度。

```
ExtendedGapAndCompliance = {
    ext_gap_compliance_id, sample_id, stage_id, extended_domain_id,
    g_ext0_vector_id, B_ext_mapping_id, free_displacement_vector_id,
    command_displacement_vector_id,
    q_ext_vector_id,
    W_struct_ext_id, C_interface_id, C_locator_id, C_joint_id,
    W_ext_matrix_id, unit_check_result,
    dependency_hash, quality_flag, metadata
}
```


8.5.4 扩展 LCP / NCP 求解对象

$$0 \leq \lambda_{\text{ext}}(s) \perp g_{\text{ext}}(s) = q_{\text{ext}}(s) + W_{\text{ext}}(s) \lambda_{\text{ext}}(s) \geq 0$$

参数含义： $\lambda_{\text{ext}}(s)$ 为扩展接触广义力，包含零件接触力、定位/卡板反力、夹持反力和单边孔边承压力； $g_{\text{ext}}(s)$ 为平衡后扩展间隙； $q_{\text{ext}}(s)$ 为扩展自由间隙； $W_{\text{ext}}(s)$ 为扩展总柔度；互补关系表示接触点分离时力为零、受压时间隙为零。

```
OverConstraintContactModel = {
    overconstraint_model_id, sample_id, stage_id, extended_domain_id,
    q_ext_vector_id, W_ext_matrix_id,
    unilateral_indices, bilateral_indices_optional,
    lambda_ext_unknown_definition, gap_sign_convention,
    solver_config_id, regularization_config_id_optional,
    fallback_policy, validity_region, metadata
}
ExtendedLCPSolution = {
    solution_id, overconstraint_model_id,
    lambda_ext, gap_ext,
    active_ext_indices, inactive_ext_indices,
    bilateral_reaction_optional,
    locator_reaction_components, clamp_reaction_components, joint_reaction_components,
    pressure_field_id_optional, local_compression_field_id_optional,
    equilibrium_residual, gap_violation, force_violation, complementarity_residual,
    active_set_stability_trace_id,
    force_nonuniqueness_report_id_optional,
    convergence_status, quality_flag, metadata
}
```

8.5.5 主动集稳定与正则化记录

$$\rho_{\chi}^k = |C^k \Delta C^{k-1}| / |C^k \cup C^{k-1}|$$

参数含义： ρ_{χ}^k 为第 k 次迭代主动集变化率； C^k 与 C^{k-1} 为相邻迭代主动集； Δ 为对称差； $\cup$ 为并集。该指标用于识别主动集振荡。

$$W_{\text{epsilon}} = W_{\text{ext}} + \text{epsilon } I$$

参数含义： W_{epsilon} 为正则化后的扩展总柔度； W_{ext} 为原扩展总柔度；epsilon 为正则化参数； I 为单位矩阵。epsilon 仅用于数值稳定，应进入日志和质量门。

```
ActiveSetStabilityTrace = {
    trace_id, solution_id, iteration_contact_sets, mode_switch_count,
    rho_chi_history, final_rho_chi, oscillation_flag,
    stabilization_action: "NONE | MERGE_POINTS | REGULARIZE_W | CHANGE_TOLERANCE | FALLBACK_SOLVER",
    quality_flag, metadata
}
LCPRregularizationConfig = {
    regularization_config_id,
    method: "NONE | EPSILON_I | DIAGONAL_SHIFT | TIKHONOV | QP",
    epsilon_value_optional, trigger_reason,
    affected_matrix_ids, output_impact_check, metadata
}
```

8.5.6 接触力非唯一报告

$$N_f = \text{null}(W_{\text{CC}}), \quad \lambda_C = \lambda_{C^*} + N_f z$$

参数含义： N_f 为主动接触子矩阵 W_{CC} 的零空间； λ_C 为主动集接触力； λ_{C^*} 为某一基准可行解； z 为零空间自由参数。若 N_f 非空，局部接触力可能存在非唯一分配。

$$\eta_y = \max_z ||H_y(\lambda_C(z) - \lambda_{C^*})|| / T_y$$

参数含义： η_y 为工程输出稳定性指标； H_y 为接触力到 KCP 或等效工程输出的灵敏度； $\lambda_C(z)$ 为零空间扰动下的接触力； T_y 为 KCP 容差或工程允许尺度。 η_y 超限表示反力非唯一会影响工程判断。

```
ForceNonuniquenessReport = {
    report_id, solution_id, active_submatrix_id, nullspace_dimension,
    nullspace_basis_id_optional,
    alternative_force_solution_summary,
    output_variation_bounds, eta_y_values,
    selected_regularization_rule,
    kcp_output_stability: "STABLE | WARN | UNSTABLE",
}
```

```
required_action, quality_flag, metadata
}
```

8.5.7 连接锁定、释放回弹与历史对象

$$\Delta x_{\text{release}} = x_{\text{release}}^{+} - T_{\text{join_to_release}} x_{\text{join}}^{+}$$

参数含义：Delta x_{release} 为释放阶段增量； x_{release}^{+} 为释放阶段后验状态； x_{join}^{+} 为连接阶段后验状态； $T_{\text{join_to_release}}$ 为从连接阶段到释放阶段的状态传递矩阵。

```
ConnectionLockHistory = {
    lock_history_id, sample_id, join_stage_id, joint_ids,
    locked_reference_dofs, locked_reference_values,
    preload_actual, joint_stiffness,
    locked_contact_mode_id, slip_history_ids_optional,
    source_ids, covariance_optional, metadata
}
ReleaseHistoryRecord = {
    release_history_id, sample_id, release_stage_id, lock_history_id,
    removed_boundary_ids, removed_load_ids, retained_joint_ids,
    residual_force_before_release, rebound_displacement, release_increment_id,
    kcp_contribution_ids, quality_flag, metadata
}
```

8.5.8 N-2-1 过程测量枚举补充

MeasurementRecord.measurement_type 建议增加 LOCATOR_REACTION、CARD_REACTION、CLAMP_HEAD_DISPLACEMENT、FASTENER_PRELOAD、HOLE_EDGE_STRAIN、CONTACT_OPEN_CLOSE_STATE、RELEASE_REBOUND、RESIDUAL_FORCE_BEFORE_RELEASE。update_target 可增加 LOCATOR_COMPLIANCE、OVERCONSTRAINT_STATE、JOINT_LOCK_HISTORY 和 RELEASE_HISTORY。

9 局部界面参数识别、柔度分解与参数库

9.1 参数识别与先验对象

$$\theta = \{C_n, C_t, \mu, \beta_r, e_{\text{proc}}\}; \quad \theta_{\text{hat}} = \underset{\theta}{\operatorname{argmin}} ||y_{\text{obs}} - y_{\text{model}}(\theta)||^2_W + \lambda ||L_{\theta}(\theta - \theta_0)||^2$$

参数含义： C_n/C_t 为局部界面柔度； μ 为摩擦系数； β_r 为回弹参数； e_{proc} 为过程误差；ParameterPrior 保存先验范围，IdentificationConfig 保存识别目标函数，ParameterEstimate 保存估计、协方差和适用域。

```
ParameterPrior = {
    parameter_prior_id, parameter_names, mean_values, covariance,
    lower_bounds, upper_bounds, physical_constraints, source, metadata
}
ParameterIdentificationConfig = {
    identification_config_id, target_parameter_names,
    identification_measurement_set_ids, local_model_ids,
    parameter_prior_id, objective_terms, weighting_matrices,
    regularization_type, regularization_weight,
    physical_constraints, solver_name, convergence_settings, metadata
}
ParameterEstimate = {
    parameter_estimate_id, parameter_names, estimated_values,
    covariance_or_confidence_intervals, correlation_matrix_optional,
    objective_value, residual_statistics, identifiability_metrics,
    active_bounds, validity_range, used_sample_ids,
    quality_flag, metadata
}
```

9.2 柔度分解与参数库

$$W_{\text{total}} = W_{\text{struct}} + C_n^{\{\text{local}\}}; \quad \Delta \gamma_c = R_n w_n + R_t d_t + R_r \Delta r$$

参数含义： W_{struct} 为结构基体柔度； $C_n^{\{\text{local}\}}$ 为未被结构模型解析的界面附加柔度； $R_n/R_t/R_r$ 为输出导向凝聚算子；InterfaceParameter 保存统计量、等效误差项和参数协方差。

```
ComplianceDecomposition = {
    compliance_decomposition_id,
    W_struct_id, Cn_local_id,
    included_physics_struct, included_physics_local,
}
```

fixture_compliance_removed, substrate_compliance_removed,
overlap_assessment, unit_consistency_check, metadata

```

}
InterfaceParameter = {
    interface_parameter_id,
    interface_id, stage_id, contact_mode_id,
    contact_area, mean_pressure, pressure_nonuniformity,
    normal_compression_rms, tangential_slip_rms_optional,
    normal_compliance_local, tangential_compliance_optional,
    equivalent_error_vector, rebound_term,
    parameter_covariance, validity_range,
    calibration_quality, source_state_ids,
    reference_state_id, increment_definition_id, metadata
}
ParameterLibrary = {
    parameter_library_id,
    interface_parameter_ids,
    contact_mode_library_id,
    indexing_variables, interpolation_rules, validity_domains,
    out_of_domain_rule,
    source_high_fidelity_case_ids, source_experiment_ids,
    library_quality_summary, metadata
}
    
```

out_of_domain_rule 不建议使用 silent_extrapolation; 推荐使用 solve_LCP_with_current_operator、request_high_fidelity_case 或 reject_prediction。

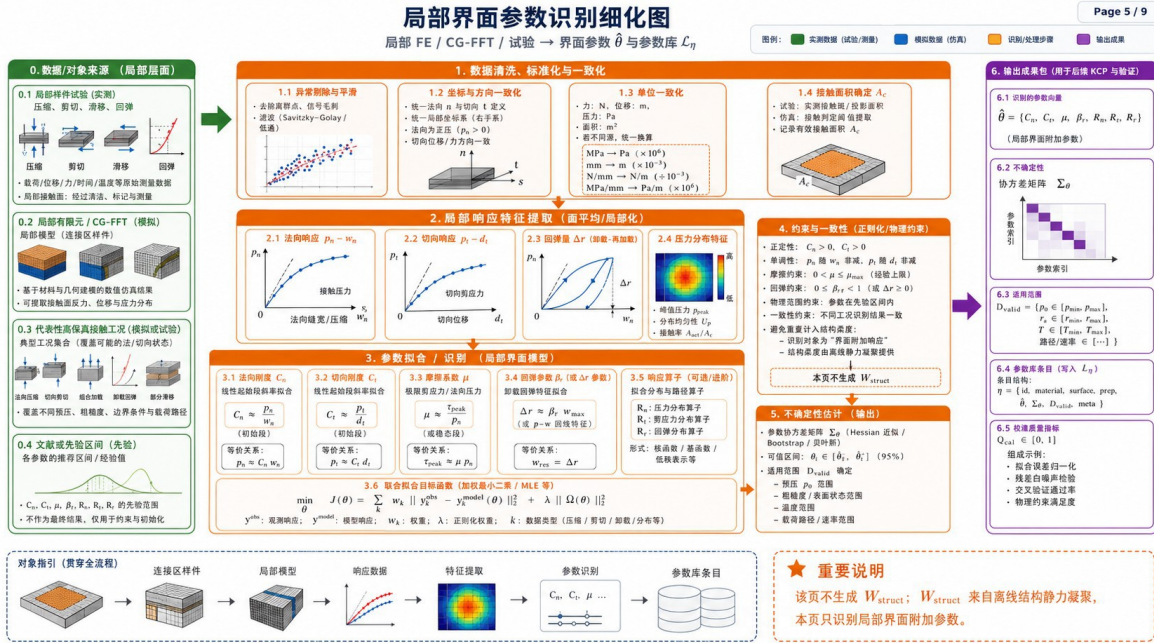


图 9-1 局部界面参数识别、约束和参数库构建

9.3 局部模型提取对象

$\Omega_{\text{local}} = \text{Gamma_a} \oplus r_margin$

参数含义: Ω_{local} 为围绕关键结合面 Gamma_a 向外扩展 r_margin 得到的局部子模型区域; r_margin 应保证切割边界远离压力峰值和位移高梯度区。

```

LocalRegionDefinition = {
    local_region_id, interface_id,
    included_geometry_regions, candidate_contact_region_ids,
    kcp_sensitive_region_ids, margin_rule, margin_distance,
    cut_boundary_definition, selection_rationale, metadata
}
    
```

```

LocalSubmodelExtraction = {
    local_model_id, global_model_id, interface_id,
    local_region_id,
    extraction_method: "geometry_cut | mesh_cut | submodel_from_global | independent_specimen_model",
    global_to_local_node_map, global_to_local_surface_map,
    retained_material_ids, retained_layup_ids_optional, retained_joint_ids,
    }
    
```

```

cut_boundary_node_ids, mesh_refinement_rule,
source_stage_ids, extraction_quality_metrics, metadata
}
BoundaryInheritanceRecord = {
  boundary_inheritance_id, local_model_id, stage_id,
  boundary_method: "DISPLACEMENT | FORCE | IMPEDANCE | MIXED",
  source_global_case_id,
  prescribed_cut_displacement_optional,
  prescribed_cut_force_optional,
  impedance_matrix_optional,
  coordinate_transform_id, interpolation_error,
  boundary_reaction_check, valid_load_range,
  quality_flag, metadata
}
LocalModelValidationResult = {
  validation_id, local_model_id,
  compared_global_case_ids,
  compared_quantities: ["cut_boundary_displacement", "reaction", "interface_displacement",
"contact_pressure"],
  error_metrics, acceptance_criteria, pass_fail, metadata
}

```

9.4 样件试验与局部响应对象

```

InterfaceTestSpecimen = {
  specimen_id, interface_id,
  side_i_material_id, side_j_material_id,
  layup_ids_optional, thickness_values,
  surface_treatment, coating_or_sealant_optional,
  fastener_or_joint_description,
  geometry_similarity_to_real_interface,
  manufacturing_batch, metadata
}
InterfaceTestSetup = {
  setup_id, specimen_id,
  fixture_id, loading_machine_id, sensor_ids,
  coordinate_system_id, support_state, load_state,
  environmental_conditions,
  fixture_calibration_id, sensor_calibration_ids,
  metadata
}
LoadingPath = {
  loading_path_id,
  stage_equivalent: "LOCATE | CLAMP | JOIN | RELEASE | LOCAL_IDENTIFICATION",
  normal_load_sequence, tangential_load_sequence_optional,
  preload_sequence_optional, hold_time_sequence_optional,
  unloading_sequence_optional, rate_information_optional, metadata
}
LocalResponseData = {
  response_id,
  source_type: "LOCAL_FE | SPECIMEN_TEST | CG_FFT | PROCESS_MEASUREMENT",
  local_model_id_optional, specimen_id_optional,
  boundary_inheritance_id_optional, loading_path_id,
  lambda_n, pressure_p_n,
  w_total, w_substrate_optional, w_interface_optional,
  tangential_slip_d_t_optional, rebound_delta_r_optional,
  strain_or_DIC_fields_optional,
  covariance_or_uncertainty,
  raw_data_reference, quality_flag, metadata
}

```

9.5 位移分解、柔度分离与参数识别对象

$$u_{\text{meas}} = u_{\text{fixture}} + u_{\text{sensor}} + u_{\text{substrate}} + u_{\text{interface}} + \text{epsilon}$$

参数含义：u_meas 为样件或局部模型总测量位移；u_fixture 为夹具柔度贡献；u_sensor 为传感器变形或偏置；u_substrate 为 CFRP/金属基体结构变形；u_interface 为界面附加响应；epsilon 为残差。只有 u_interface 应用于识别局部附加柔度。

```

DeformationDecompositionResult = {
  decomposition_id, response_id,
  fixture_component, sensor_component, substrate_component, interface_component,
  removed_deformation_source_ids, remaining_deformation_source_ids,
  decomposition_method: "calibration_subtraction | local_FE_subtraction | joint_identification |

```

```

bayesian_decomposition",
  component_covariances, identifiability_metrics, quality_flag, metadata
}
ParameterIdentificationConfig += {
  local_response_data_ids,
  deformation_decomposition_ids,
  target_discretization: "scalar | zonal_scalar | diagonal | block | full_matrix | nonlinear_piecewise",
  area_weight_matrix_id,
  generalized_force_conversion_rule,
  boundary_inheritance_ids,
  data_sufficiency_rule,
  forbidden_flexibility_source_ids,
  metadata
}
ParameterEstimate += {
  source_local_response_data_ids,
  source_boundary_inheritance_ids,
  decomposition_result_ids,
  estimated_Cn_local,
  estimated_Ct_optional,
  estimated_mu_optional,
  estimated_beta_r_optional,
  load_segment_breakpoints_optional,
  posterior_covariance,
  parameter_correlation_matrix,
  validity_domain_id,
  metadata
}
ComplianceSeparationReport = {
  separation_report_id,
  W_struct_id, Cn_local_id,
  included_physics_struct, included_physics_local,
  removed_fixture_compliance, removed_sensor_compliance, removed_substrate_compliance,
  overlap_source_ids,
  overlap_assessment: "PASS | WARN | FAIL",
  unit_consistency_check, area_weight_conversion_check,
  recommended_action, metadata
}

```

9.6 参数库对象字段补充与适用域

```

ParameterLibrary += {
  local_model_ids, local_region_ids, boundary_inheritance_ids,
  specimen_ids, loading_path_ids,
  compliance_separation_report_ids,
  indexing_variables: ["material", "layup", "surface", "pressure", "preload", "temperature",
"boundary_method", "contact_mode"],
  out_of_domain_rule: "request_high_fidelity_case | reject_prediction | solve_LCP_with_current_operator",
  library_quality_summary, metadata
}
ValidityDomain = {
  validity_domain_id,
  material_range,
  layup_or_thickness_range,
  surface_state_range,
  normal_pressure_range,
  tangential_load_range_optional,
  preload_range_optional,
  temperature_range_optional,
  contact_mode_range,
  boundary_method_range,
  loading_path_range,
  geometry_scale_range,
  metadata
}

```

10 过程实测状态/参数更新与校准对象

10.1 过程观测模型与状态更新

$$z_meas^{\wedge}(s)=h^{\wedge}(s)(x^{\wedge}(s),\theta,e_proc)+v^{\wedge}(s); \quad x_hat^{\wedge}(s),\theta_hat = \operatorname{argmin} ||z_meas-h(x,\theta)||^2_{\{R^{-1}\}} + ||x-x_pred||^2_{\{P^{-1}\}} + ||\theta-\theta_0||^2_{\{\Sigma^{-1}\}}$$

参数含义：z_meas 为阶段过程测量； $x^s(s)$ 为阶段状态；theta 为可校准参数；R/P/Sigma 分别表达测量、状态先验和参数先验不确定度；StateUpdateResult 保存更新前后残差和模式变化。

```
ProcessObservationModel = {
  observation_model_id, stage_id,
  measured_feature_ids, state_variable_ids, parameter_ids_optional,
  H_z_or_function_reference,
  bias_vector_optional,
  covariance_R_z,
  measurement_coordinate_rule,
  validity_region, metadata
}
StateUpdateConfig = {
  state_update_config_id,
  stage_id,
  predicted_state_id,
  process_measurement_set_ids,
  observation_model_id,
  prior_state_covariance_id_optional,
  update_method,
  regularization_weight,
  complementarity_constraints_enabled,
  updated_state_field_names,
  solver_config, metadata
}
StateUpdateResult = {
  state_update_result_id,
  predicted_state_id,
  updated_state_id,
  used_measurement_ids,
  predicted_measurements,
  residual_before, residual_after,
  posterior_covariance_optional,
  contact_mode_changed,
  constraint_residuals,
  updated_target_summary,
  quality_flag, metadata
}
```

10.2 模型校准对象

$$\theta_{\text{hat}} = \underset{\theta}{\operatorname{argmin}} \left(\sum_s \left\| z_{\text{meas}}^s - h^s(s)(\theta_{\text{hat_SMS}}, \theta) \right\|^2_{R_s^{-1}} + \lambda_{\theta} \left\| L_{\theta}(\theta - \theta_0) \right\|^2 \right), \quad \text{sample not in } D_{\text{validate}}$$

参数含义：calibration_sample_ids 只能来自 D_calibrate；forbidden_validation_sample_ids 用于程序层面禁止验证集回流调参。

```
ModelCalibrationConfig = {
  calibration_config_id,
  calibrated_parameter_names,
  calibration_sample_ids,
  stage_ids,
  sms_update_result_ids,
  measurement_set_ids,
  model_components,
  loss_function,
  regularization,
  bounds_and_constraints,
  cross_validation_rule,
  forbidden_validation_sample_ids,
  metadata
}
```

10.3 更新决策、参数传播、重求与回退对象

```
UpdateDecisionRecord = {
  update_decision_id, sample_id, stage_id,
  residual_signature, candidate_update_targets,
  selected_update_targets, frozen_targets,
  identifiability_metrics,
  update_permission_check,
  evidence_measurement_ids,
  rationale, quality_flag, metadata
}
```

```
ParameterPropagationRule = {
  propagation_rule_id, parameter_name_or_group,
  source_scope, target_scope,
  allowed_from_stage_ids, allowed_to_stage_ids,
  persistence_policy,
  validity_domain_requirement,
  transformation_or_reset_rule,
  covariance_propagation_rule,
  out_of_domain_action, metadata
}
ReSolveRequirement = {
  resolve_requirement_id, update_result_id,
  changed_q, changed_W, changed_K_or_B,
  contact_mode_changed,
  required_level: "NONE | RESOLVE_LCP | REBUILD_W | REBUILD_IRED | HIGH_FIDELITY",
  invalidated_cache_ids, reason, metadata
}
UpdateRollbackRecord = {
  rollback_id, failed_update_result_id,
  restored_state_id, restored_parameter_snapshot_ids,
  failure_reason, fallback_action,
  retained_measurement_ids, metadata
}
```

对象	允许更新	传播	重求级别
alpha_sms / delta_sms	PRE_ASSEMBLY_REFERENCE_MEASUREMENT	READ_ONLY_ALL_STAGES	变化则 REBUILD_G0 + 全流程
pose / fixture_bias / actual_load	对应阶段过程测量	STATE_INITIAL_CONDITION / LOAD_INCREMENT	RESOLVE_LCP
Cn / Ct / mu	CALIBRATE 且可辨识、D_valid 内	INTERFACE_CONDITIONAL	REBUILD_W / 摩擦 NCP
joint_stiffness / locked_reference	JOIN	PERSIST_TO_RELEASE	REBUILD_IRED 或 RESOLVE_LCP
beta_r / Rr	RELEASE_CALIBRATE	FINAL_OUTPUT_ONLY	重算释放输出
locator_compliance / overconstraint_state	LOCATE/CLAMP 过程测量且可辨识	OVERCONSTRAINT_CONDITIONAL	REBUILD_W_EXT 或 RESOLVE_LCP

11 界面等效误差项、KCP 预测与贡献分解

11.1 等效误差 J-T 传递

$$\Delta \gamma_{\{c,a\}^{\{s\}}} = R_n w_{\{n,a\}^{\{s\}}} + R_t d_{\{t,a\}^{\{s\}}} + R_r \Delta r_a^{\{s\}};$$
$$\Delta y = J_{\text{part}} \Delta q + J_{\text{interface}} \Delta \gamma + J_{\text{proc}} e_{\text{proc}}$$

参数含义：Delta gamma 为界面等效误差向量；w_n/d_t/Delta r 为法向压缩、切向滑移和回弹；J_{part}/J_{interface}/J_{proc} 为小位移、界面项和过程误差到 KCP 的传递矩阵。

```
EquivalentErrorItem = {
  equivalent_error_id,
  interface_id, stage_id, contact_mode_id,
  error_vector,
  components: {
    normal_compression_contribution,
    tangential_slip_contribution_optional,
    rebound_contribution_optional
  },
  mapping_operators: {Rn_id, Rt_id_optional, Rr_id_optional},
  covariance_optional,
  reference_state_id,
  increment_definition_id,
  source_state_ids,
  quality_flag, metadata
}
JTTransferModel = {
  jt_model_id,
  kcp_ids,
  part_pose_source_ids,
  interface_error_source_ids,
  J_part_matrices,
  J_interface_matrices,
  J_process_matrix,
  linearization_state,
```

```
validity_range, metadata
}
```

11.2 单样本预测结果 KCPPredictionResult

```
Delta y_KCP = J_SMS delta_hat_SMS + Sigma_s Sigma_a J_a^(s) Delta gamma_{c,a}^{(s)} + J_proc
e_proc
```

参数含义：sms_contribution 为装配前几何贡献；contact/slip/rebound_contribution 为接触压缩、滑移和回弹贡献；process_contribution 为工艺误差贡献；contribution_reference_state_ids 用于检查各贡献参考状态是否一致。

```
KCPPredictionResult = {
  prediction_result_id,
  prediction_input_id,
  kcp_ids,
  predicted_values,
  nominal_values,
  tolerance_status,
  sms_contribution,
  contact_contribution,
  slip_contribution_optional,
  rebound_contribution,
  process_contribution,
  measurement_uncertainty_contribution_optional,
  contribution_reference_state_ids,
  contact_mode_ids,
  out_of_domain_flags,
  runtime_breakdown,
  convergence_summary,
  quality_flag, metadata
}
```



图 11-1 界面接触状态统计与输出导向等效凝聚

11.3 变形贡献台账与重复计入检查

Source(SMS) n Source(StageIncrement^(s)) = empty; 同一 KCP 路径内 (source_id, increment_definition_id) 只能消一次

```
ContributionRecord = {
  contribution_id, sample_id, target_kcp_id,
  source_class: "SMS | STAGE_INCREMENT | PROCESS_ERROR | REBOUND | MEASUREMENT_UNCERTAINTY |
  OVERCONSTRAINT_REACTION",
  source_id, origin_stage_id_optional,
  reference_state_id, increment_definition_id,
  absolute_or_increment,
  included_deformation_source_ids,
```

```

excluded_deformation_source_ids,
contribution_vector, covariance_optional,
consumed_by_prediction_id, quality_flag, metadata
}
DeformationContributionLedger = {
contribution_ledger_id, sample_id,
contribution_record_ids,
allowed_source_classes,
aggregation_policy,
unique_source_key_rule,
parent_ledger_id_optional,
ledger_hash, quality_flag, metadata
}
DoubleCountCheckResult = {
double_count_check_id, contribution_ledger_id,
duplicate_source_ids,
reference_state_conflicts,
absolute_increment_conflicts,
included_excluded_overlap,
check_rules_version,
pass_warn_fail, required_action, metadata
}
KCPPredictionResult += {
contribution_ledger_id,
double_count_check_result_id,
aggregation_policy: "FINAL_ABSOLUTE_STATE | SUM_STAGE_INCREMENTS",
source_consumption_summary,
reference_state_consistency_flag
}

```

检查规则	失败示例
唯一变形源	测量夹持变形既包含在 SMS，又作为 CLAMP 增量加入。
绝对/增量互斥	同时累加 JOIN 绝对状态与 RELEASE 绝对状态。
参考态一致	SMS 相对 FREE，阶段增量相对未知测量态且无转换记录。
包含/排除源不重叠	同一 source_id 同时出现在 included_sources 和 excluded_sources。
过约束源唯一	定位反力既作为 LocatorElement 柔度补偿又作为 StageIncrement 贡献时未分清源。

KCP 误差传递与 Monte Carlo 外循环细化图

单样本快速预测 + 外层统计抽样

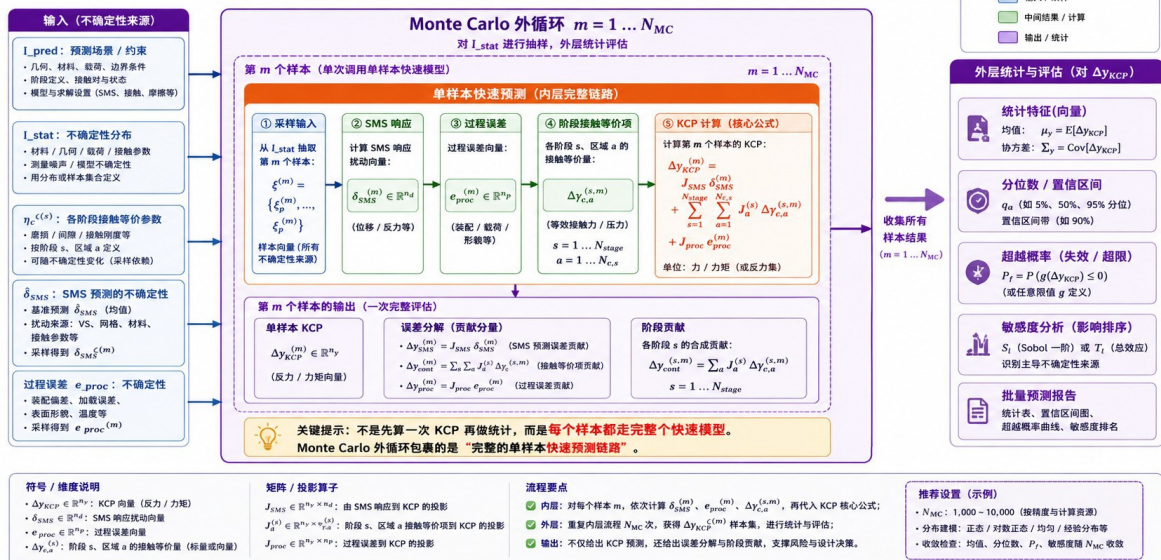


图 11-2 KCP 误差传递与 Monte Carlo 外循环

12 Monte Carlo、敏感性、验证与高保真对比

12.1 统计配置和随机样本

$\mathbf{x}^{(m)} \sim D_{\mathbf{x}}$, $\mathbf{x}_i = [\alpha, e_{\text{proc}}, \theta, \epsilon_{\text{meas}}]$; 第 m 个样本调用 F_{fast} 得到 $\Delta y_{\text{KCP}}^{(m)}$

参数含义：joint_distribution_id 定义联合分布；RandomSample 保存每次抽样的 SMS、界面参数和过程误差；correlation_preserved 说明是否保留输入相关性。N-2-1 情况下 $\mathbf{x}_i$ 还可包含 locator_compliance_sample、command_displacement_sample 和 preload_sample。

```
MonteCarloConfig = {
    mc_config_id,
    random_variable_ids,
    joint_distribution_id,
    sample_count,
    random_seed,
    sampling_method,
    batch_size,
    parallel_settings,
    mode_library_usage,
    high_fidelity_fallback_budget,
    stopping_rule_optional, metadata
}
RandomSample = {
    random_sample_id,
    mc_config_id,
    sample_index,
    alpha_sms,
    interface_parameter_sample,
    process_error_sample,
    measurement_error_sample_optional,
    locator_compliance_sample_optional,
    preload_or_command_displacement_sample_optional,
    correlation_preserved,
    source_distribution_ids, metadata
}
```

12.2 统计、敏感性与验证结果

$\mu_y = 1/N \sum \mathbf{y}^{(m)}$, $\Sigma_y = 1/(N-1) \sum (\mathbf{y}^{(m)} - \mu_y)(\mathbf{y}^{(m)} - \mu_y)^T$, $P_f = 1/N \sum \mathbb{1}[\mathbf{y}^{(m)} \notin [L, U]]$, $\mathbf{r}_{\text{val}} = \mathbf{y}_{\text{val}} - \mathbf{y}_{\text{pred}}$

参数含义：mean/covariance/quantiles 描述 KCP 分布；failure_probabilities 表示超差概率；SensitivityResult 记录输入贡献；ValidationResult 记录独立验证残差和通过/失败结论。

```
StatisticalPredictionResult = {
    statistical_result_id,
    mc_config_id,
    kcp_ids,
    mean_vector,
    covariance_matrix,
    quantiles,
    failure_probabilities,
    confidence_intervals,
    mode_frequency_statistics,
    out_of_domain_rate,
    nonconvergence_rate,
    overconstraint_mode_frequency_optional,
    runtime_statistics, metadata
}
SensitivityResult = {
    sensitivity_result_id,
    target_kcp_ids,
    input_variable_groups,
    method,
    local_mode_sensitivities_optional,
    first_order_indices,
    total_order_indices_optional,
    ranking,
    confidence_intervals,
    sample_ids, metadata
}
```



```

}
ValidationResult = {
    validation_result_id,
    model_version,
    validation_sample_ids,
    reference_type,
    kcp_ids,
    predicted_values,
    reference_values,
    residuals,
    MAE, RMSE, maximum_absolute_error,
    correlation_optional,
    failure_probability_error_optional,
    contact_mode_agreement_optional,
    overconstraint_reaction_agreement_optional,
    acceleration_ratio_optional,
    acceptance_criteria,
    pass_fail, metadata
}

```



图 12-1 独立验证、适用域和升级触发流程

13 质量门、日志、缓存失效与操作交付物

13.1 质量门 QualityGate

$Q=PASS \iff \text{for all } j, c_j(x) \leq \tau_j$; $Q=WARN/FAIL$ 由超限程度和 $action_required$ 判定

参数含义: $check_items$ 为检查项; $metric_value$ 为计算值; $threshold_rule$ 为阈值规则; $pass_fail$ 为质量门结论; QualityGate 将技术说明书的操作卡落实为可执行检查。

```

QualityGate = {
    quality_gate_id,
    gate_name,
    target_object_type,
    target_object_ids,
    check_items,
    thresholds,
    pass_fail,
    warn_items,
    fail_items,
    reviewer_optional,
    timestamp, metadata
}

CheckItem = {
    check_name,
    metric_name,

```

```
metric_value,  
unit_optional,  
threshold_rule,  
result: "PASS | WARN | FAIL",  
action_required_optional  
}
```

环节	建议质量门	失败处理
输入治理	坐标闭合、单位一致、缺失值、用途标签、数据分组。	重新配准、剔除或降级。
SMS 更新	H_m N 秩/条件数、残差、后验方差、基外误差。	减少模态、增加测点、修正基准或扩充 N。
接触域	法向/位序、面积闭合、点质量、候选域覆盖。	扩大候选域、重建对应、更新法向。
结构凝聚	矩阵秩、对称性、半正定、条件数、全阶-凝聚误差。	修正约束、重建 K_Gamma 或回到高保真。
互补求解	平衡、间隙、接触力、互补残差、主动集稳定。	调整缩放/求解器、回算、扩展模式库。
局部模型提取	局部域覆盖压力峰值和 KCP 敏感区；切割边界远离高梯度区；映射关系完整。	扩大局部域、重建局部网格或回到全局高保真。
边界继承	位移/反力/阻抗边界误差满足阈值；边界反力与全局模型一致。	改用阻抗边界或扩大局部域。
样件/位移分解	夹具/传感器校准有效；u_fixture、u_sensor、u_substrate 可分离。	重测、降级或增加多载荷、多位置数据。
柔度分离	W_struct 与 Cn_local 的 included_source_ids 不重叠；单位和 Q_A 转换正确。	扣除重复柔度、重做识别或标记 FAIL。
N-2-1 过约束	W_ext 对称/半正定、rho_chi 稳定、反力非唯一不影响 KCP、locator/joint 源不重复。	MERGE_POINTS、REGULARIZE_W、FALLBACK_SOLVER、HIGH_FIDELITY 或 REJECT。
KCP 预测	参考状态一致、量纲、防重复贡献、模式可行。	重定义贡献项或重求接触。
验证	D_validate 独立性、MAE/RMSE、超差率误差、适用域。	重新发布、警告、拒绝或新版本。

13.2 运行日志与模型发布

```
M_v = {I, theta_hat, L_eta, code, Q, V},    release(M_v) <=> ValidationResult=PASS 且  
QualityGate 全部通过
```

参数含义：RunLog 记录一次运行的输入、缓存、警告、错误和输出；ModelVersionRelease 固化模型版本、适用域、识别/校准/验证数据摘要和再验证条件。

```
RunLog = {  
  run_id, model_version, input_package_id,  
  started_at, finished_at, operator_or_script,  
  software_versions,  
  object_ids_used,  
  cache_ids_used,  
  quality_gate_ids,  
  warnings,  
  errors,  
  runtime_summary,  
  output_object_ids, metadata  
}  
ModelVersionRelease = {  
  model_version,  
  released_at,  
  released_by,  
  valid_domain,  
  frozen_object_ids,  
  training_identification_data_summary,  
  calibration_data_summary,  
  validation_result_ids,  
  known_limitations,  
  required_revalidation_conditions, metadata  
}
```

14 端到端对象流、伪代码与实施清单

14.1 离线建库阶段对象流

```
OfflineBuild: (I0, I_key, I_Gamma, I_stage, D_identify) -> (I_red, ParameterLibrary,  
PropagationRules, Q_offline)
```

```

I0 = build_nominal_model(parts, interfaces, materials, topology)
I_key = define_KCM_KCP(features, tolerances, directions)
I_Gamma = build_contact_domains(I0, I_key)
I_stage = build_stage_inputs(process_plan, boundaries, loads)
I_red = condense_structure(I0, I_Gamma, I_stage)
for each key interface:
    local_region = define_local_region(interface, KCP_sensitivity, margin_rule)
    local_model = extract_local_submodel(global_FE, local_region)
    boundary_records = inherit_boundaries(global_FE_cases, local_model, stages)
    validate_local_model(local_model, boundary_records)
    test_specs = design_interface_tests(material, surface, loading_path)
    responses = collect_local_FE_CGFFT_and_test_response(local_model, test_specs)
    decompositions = decompose_deformation(responses, fixture_calibration, sensor_calibration,
    substrate_FE)
    theta = identify_interface_parameters(decompositions)
    compliance_report = check_compliance_separation(I_red.W_struct, theta.Cn_local)
    ParameterLibrary.register(theta, compliance_report, D_valid)
for each overconstraint group:
    build_locator_elements_and_compliance(group)
    validate_extended_domain(group, I_Gamma, I_stage)
PropagationRules = define_stage_parameter_and_transition_rules()
QualityGate.run_all([I0, I_key, I_Gamma, I_stage, I_red, ParameterLibrary, OverConstraintDomains])

```

14.2 单样本在线预测对象流

```

meas = load_measurements(sample_id)
ms = build_measurement_state_snapshot(meas.part_kcm)
comp = compensate_measurement_state(ms, fixture_model, measured_loads)
kcm_ref = build_reference_kcm(ms, comp)
sms_update = update_SMS(kcm_ref, SMSPrior, SMSBasis, KCMDefinition)
sms_pkg = build_sms_transfer_package(sms_update, comp)
gap = build_gap_field(sms_pkg, ContactDomain, SMSPointCloudMapping)
previous_pkg = initialize_stage_package(gap)
for stage in stages:
    K_Gamma = load_condensed_operator(I_red[stage], material_boundary_hash)
    F_Gamma = build_reduced_force_vector(I_stage[stage], actual_loads)
    if overconstraint_enabled(stage):
        ext = build_extended_gap_and_compliance(gap, K_Gamma, F_Gamma, LocatorGroups[stage])
        model = build_overconstraint_contact_model(ext)
        state = solve_extended_complementarity(model, warm_start=previous_pkg.mode)
    else:
        pred = stage_predict(previous_pkg, I_red[stage], I_stage[stage], ParameterLibrary)
        state = solve_complementarity(pred, warm_start=previous_pkg.mode)
    trace = build_contact_computation_trace(gap, state.q, state.W, dependencies)
    trace.attach_solution(state)
    if process_measurement_available(stage):
        decision = diagnose_update_target(state, process_meas[stage])
        state = constrained_update_and_resolve(state, decision, process_meas[stage])
    increment = build_stage_increment(previous_pkg, state)
    ledger.register(increment)
    previous_pkg = build_stage_posterior_package(state, increment)
check_no_double_count(ledger)
eta = condense_interface_error(stage_packages)
pred = predict_KCP(sms_pkg, eta, I_pred, ledger)

```

14.3 Monte Carlo 批量对象流

```

samples = draw_random_samples(I_stat, sms_update_cov, parameter_cov, process_error_cov)
for sample in samples:
    pred_m = run_single_sample_fast_model(sample)
    collect(pred_m)
stat_result = summarize_statistics(predictions)
sensitivity = compute_sensitivity(predictions, samples)
validation = compare_with_D_validate(stat_result, validation_set)

```

14.4 防重复计算的实现约束

- 每次 run_single_sample_fast_model 必须创建独立 ContributionLedger。
- 阶段更新不得修改 SMSUpdateTransferPackage。
- 任何 q/W/K/模式变化必须由 ReSolveRequirement 驱动缓存失效和重求。
- N-2-1 扩展对象的 locator/joint/clamp 源必须进入同一 source ledger，避免同一定位器变形既作为柔度又作为阶段贡献重复计入。

- ContactComputationTrace 至少记录 g0、q、W、lambda、p/w、主动集和残差，保证 KCP 可回放。

附录 A 枚举值、字段命名和最小可运行数据包

A.1 关键枚举值

枚举字段	推荐取值
support_state	FREE、KNOWN_SUPPORT、FIXTURED、UNKNOWN
load_state	FREE、LOCATED、CLAMPED、JOINED、RELEASED、DISASSEMBLED_FREE
compensation_method	FE_CORRECTION、MULTI_POSE_SEPARATION、STATE_DIFFERENCE、NONE_REJECT
update_target	SMS、POSE、STAGE_STATE、PROCESS_LOAD、INTERFACE_PARAMETER、REBOUND_PARAMETER、KCP_VALIDATION、MONITOR_ONLY、LOCATOR_COMPLIANCE、OVERCONSTRAINT_STATE、JOINT_LOCK_HISTORY、RELEASE_HISTORY
measurement_type	point_cloud、KCM、force、gap、strain、KCP、LOCATOR_REACTION、CARD_REACTION、CLAMP_HEAD_DISPLACEMENT、FASTENER_PRELOAD、HOLE_EDGE_STRAIN、CONTACT_OPEN_CLOSE_STATE、RELEASE_REBOUND、RESIDUAL_FORCE_BEFORE_RELEASE
data_role	IDENTIFY、CALIBRATE、VALIDATE、MONITOR、EXCLUDE
boundary_method	DISPLACEMENT、FORCE、IMPEDANCE、MIXED
source_type	LOCAL_FE、SPECIMEN_TEST、CG_FFT、PROCESS_MEASUREMENT
decomposition_method	calibration_subtraction、local_FE_subtraction、joint_identification、bayesian_decomposition
target_discretization	scalar、zonal_scalar、diagonal、block、full_matrix、nonlinear_piecewise
group_type	N21、MULTI_POINT、CARD、CLAMP、JOINT
element_type	LOCATOR_POINT、SUPPORT_PAD、CARD_LINE、CLAMP_HEAD、FASTENER_BEARING、JOINT_LOCK
required_level	NONE、RESOLVE_LCP、REBUILD_W、REBUILD_IRED、HIGH_FIDELITY
quality_flag	PASS、WARN、FAIL、REJECT、OUTLIER_REVIEWED、MISSING

A.2 最小可运行数据包

P_min={I0,I_key,I_meas,I_Gamma,I_stage,I_red,L_eta,M_LCP,M_pred,Transition,Ledger,V}

```
MinimumRunnablePackage_V2_5 = {
  I0: {Part, Interface, NominalModel, Material, AssemblyTopology, SMSPrior, LocalRegionDefinition},
  I_key: {KCMDefinition, KCPDefinition, FeatureMapping},
  I_meas: {MeasurementPackage, MeasurementRecord, DataSplit,
    MeasurementStateSnapshot, MeasurementStateCompensationResult,
    InterfaceTestSpecimen, InterfaceTestSetup, FixtureCalibration, SensorCalibration,
    LoadingPath},
  sms_update: {SMSUpdateConfig, SMSUpdateResult, SMSUpdateTransferPackage, SMSPointCloudMapping},
  I_Gamma: {ContactDomain, ContactPoint, GapField, AreaWeightMatrix, G, Bn,
    ExtendedContactDomain, ExtendedGapAndCompliance},
  I_stage: {StageInput, BoundaryItem, LoadItem, ReducedForceVector,
    OverConstraintLocatorGroup, LocatorElement, LocatorCompliance,
    StageTransitionRecord, StageStateSnapshot, StageParameterSnapshot,
    StageIncrement, StagePosteriorPackage, ConnectionLockHistory, ReleaseHistoryRecord},
  I_red: {DOFPartition, CondensedOperator, W_struct, FactorizationCache,
    LocalSubmodelExtraction, BoundaryInheritanceRecord, LocalModelValidationResult},
  parameter_library: {LocalResponseData, DeformationDecompositionResult,
    ParameterEstimate, ComplianceSeparationReport,
    InterfaceParameter, ParameterLibrary, ValidityDomain},
  solver: {InterfaceLCPModel, ComplementaritySolverConfig, ContactComputationTrace,
    OverConstraintContactModel, ExtendedLCPSolution,
    ActiveSetStabilityTrace, LCPRegularizationConfig, ForceNonuniquenessReport},
  prediction: {EquivalentErrorItem, JTTTransferModel, ContributionLedger,
    DoubleCountCheckResult, KCPPredictionResult},
  validation: {ValidationResult, QualityGate, RunLog, ModelVersionRelease}
}
```

A.3 与技术说明书章节对应速查

技术说明书章节	本数据结构说明书对象
第 3 章: KCM-SMS 与参考态	MeasurementStateSnapshot、MeasurementDeformationEstimate、

	MeasurementStateCompensationResult、SMSUpdateTransferPackage
第 4-5 章：接触域与结构凝聚	ContactDomain、GapField、SMSPointCloudMapping、CondensedOperator、ReducedForceVector、FactorizationCache
第 6 章：局部界面参数识别	LocalRegionDefinition、LocalSubmodelExtraction、BoundaryInheritanceRecord、InterfaceTestSpecimen、LocalResponseData、DeformationDecompositionResult、ComplianceSeparationReport、ValidityDomain
第 7 章：多阶段互补与 N-2-1 过约束	InterfaceLCPModel、LCPSolution、StageStateSnapshot、OverConstraintLocatorGroup、ExtendedContactDomain、OverConstraintContactModel、ExtendedLCPSolution、ActiveSetStabilityTrace、ForceNonuniquenessReport、ConnectionLockHistory、ReleaseHistoryRecord
第 8 章：过程实测更新	ProcessObservationModel、StateUpdateConfig、StateUpdateResult、UpdateDecisionRecord、ReSolveRequirement、UpdateRollbackRecord
第 9-10 章：界面凝聚与 KCP	EquivalentErrorItem、ContributionRecord、DeformationContributionLedger、DoubleCountCheckResult、KCPPredictionResult
第 11-12 章：统计与验证	MonteCarloConfig、StatisticalPredictionResult、SensitivityResult、ValidationResult、ModelVersionRelease

使用说明

本说明书不要求一次性实现所有对象。硕士论文阶段可优先实现最小可运行数据包；若进入程序开发或工程试验阶段，再逐步扩展参数库、模式库、版本治理和质量门对象。任何新增字段都应保持两个原则：能够追溯来源，能够说明它相对于哪个参考状态定义。